

Spectral-Local Residual Networks for Wave Inverse Reconstruction

May 6, 2026

Abstract

Wave inverse problems like coherent diffractive imaging (CDI) require networks to combine global measurement-domain coupling with sharp real-space synthesis. Pure-convolutional and pure-spectral models each resolve this tension only partially. We show that closing this gap on simulated CDI requires getting two choices right at once: training through the known diffraction model, and placing spectral mixing in the encoder rather than the bottleneck of an otherwise local architecture. Switching from object-space supervision to a PtychoPINN-style measurement-consistency loss reduces FFNO’s amplitude MAE more than fourfold ($0.35 \rightarrow 0.08$), with consistent gains across every architecture we train both ways. Holding that loss fixed, we introduce **SRU-Net**, which interleaves truncated Fourier mixing with local convolution in each encoder block, processes the result through a six-block residual bottleneck, and decodes with transposed convolutions. SRU-Net cuts CDI amplitude MAE $3.0\times$ over FFNO and $4.6\times$ over a strong CNN baseline (0.027 vs. 0.082 vs. 0.123) and reaches amplitude SSIM 0.99. Ablations indicate the gain depends on where spectral mixing is placed, not on spectral capacity alone: replacing the residual bottleneck with FFNO-style spectral blocks or weakening the decoder regresses both error metrics. The same design transfers to Born–Rytov diffraction tomography, where SRU-Net reaches relative L_2 of 0.288 against 0.451 for FFNO and 0.380 for a model-based Born inverse. On forward compressible Navier–Stokes prediction, FFNO remains strongest; SRU-Net is second by aggregate error, while U-NO is close to FNO in aggregate error and stronger at higher frequencies.

1 Introduction

Problems such as phase retrieval, inverse scattering, and forward PDE prediction combine long-range coupling with localized structure. In lensless computed imaging, each detector measurement depends on the target object and probe interaction through global wave propagation, while the reconstruction is a real-space object with localized amplitude and phase structure. In fluid dynamics, large-scale coherent dynamics coexist with localized features such as shocks and vortices. Across these settings, learned models must propagate information globally without erasing localized spatial structure.

Classical forward solvers often require fine discretization, while inverse wave reconstruction commonly relies on iterative physics-based reconstruction or projection algorithms. Standard computer vision architectures adapted to physics-prediction tasks provide fast approximate solutions, but their accuracy is limited in domains with prominent scale mixing. The accuracy gap reflects a specific architectural mismatch: architectures that are efficient on natural images, such as CNNs, process local spatial neighborhoods well but lack direct global receptive fields for long-range physical interactions.

Fourier neural operator (FNO) networks address the opposite side of this tradeoff. They are effective models for continuous physical systems because spectral layers provide global mixing [2].

However, FNO-style models retain a finite set of Fourier modes, and this truncation can bias predictions toward lower-frequency structure and weaken sharp localized features. Recent extensions mitigate this limitation by coupling global Fourier layers with local or multiscale operators [4, 3]. These convolutional, U-Net, or localized-kernel branches help preserve fine-scale structure while retaining long-range mixing. Such hybrids improve some PDE benchmarks, but they are not uniformly better than strong spectral-only architectures such as the factorized Fourier neural operator [5]. We focus on convolutional and spectral models; attention-based operators such as Galerkin Transformer and related attention-operator hybrids are a separate model family outside the scope of this study.

Coherent diffractive imaging (CDI) is our primary test case because it stresses both sides of the global-local problem. Diffraction measurements are globally coupled by the wave forward model, while the desired output is a localized complex object. The inverse map must therefore interpret measurement-domain structure across the full detector field and synthesize a sharp real-space reconstruction. Our experiments show that FNO-based architectures used in forward physical prediction can struggle to synthesize accurate real-space images from diffraction measurements, although they still improve over purely convolutional models in several comparisons.

For CDI, the training objective is as important as the architecture. We do not propose a new CDI loss; instead, we use the established PtychoPINN-style physics-consistency objective, in which a predicted object is passed through known real-space consistency and diffraction operators and the loss is applied in measurement space [1]. This objective is not a minor implementation detail in our experiments. Models trained with supervised loss show meaningfully worse aggregate reconstruction quality than the corresponding physics-consistency-trained models across the tested architectures, with the clearest gaps in amplitude reconstruction. We therefore use the same physics-consistency objective for the CDI architecture comparison and ask which network works best once the standard CDI physics constraint is imposed.

We introduce SRU-Net, a spectral-convolutional architecture for wave inverse reconstruction. Each encoder block combines a truncated Fourier mixing branch with a local convolutional branch inside a residual update. The Fourier branch lets the model aggregate information across the full measurement field, while the convolutional branch preserves grid-aligned local features. A residual bottleneck processes the mixed representation at reduced spatial resolution, and a convolutional decoder synthesizes the real-space output. The architecture is designed to separate the inverse-mapping role of global spectral mixing from the image-synthesis role of local convolutional decoding.

Our contributions are threefold. *First*, we argue that for wave inverse reconstruction, spectral mixing belongs in the encoder, paired with a local residual bottleneck and a convolutional decoder; we instantiate this as SRU-Net. *Second*, on synthetic CDI under a shared PtychoPINN-style physics-consistency objective, SRU-Net cuts amplitude MAE to 0.027, a $3.0\times$ improvement over FFNO (0.082) and $4.6\times$ over a CNN baseline (0.123), and reaches amplitude SSIM 0.99. The same objective reduces amplitude error for every architecture trained both ways, isolating the architectural gap from the loss. *Third*, ablations isolate the placement: the convolutional decoder is load-bearing (replacing it with a linear decoder drives amplitude error to $5\times$ the SRU-Net value), and the local residual bottleneck cannot be replaced by spectral blocks without regressing both amplitude and phase metrics.

2 Related Work

2.1 Operator learning and spectral-local architectures

Neural operators learn mappings between fields and are widely used for scientific machine learning. Fourier neural operators (FNOs) apply global spectral convolutions to represent solution operators for parametric PDEs [2]. Factorized Fourier neural operators (FFNOs) make this parameterization deeper and more efficient by factorizing Fourier transforms, sharing kernel integral operators across layers, and using residual temporal updates [5]. Their global receptive field is attractive for wave propagation, diffraction, and fluid dynamics, where distant regions of a field are coupled by the governing physics. PDEBench provides a controlled setting for evaluating such models on forward physical prediction tasks, including compressible Navier–Stokes dynamics [6].

Pure spectral models provide direct global coupling, but Fourier-mode truncation can bias predictions toward smoother, lower-frequency structure. This is problematic when the target contains sharp local features, such as object boundaries in inverse imaging or high-gradient structures in flow. Spectral-local architectures address this limitation by combining Fourier layers with convolutional, U-Net, or localized-kernel branches [4, 3]. These models motivate the SRU-Net design, but they do not settle which architectural components are needed for physics-consistent wave inverse reconstruction.

2.2 Physics-constrained wave inverse reconstruction

Hoidn, Mishra, and Mehta introduced PtychoPINN for unsupervised scanning-CDI reconstruction [1]. The method learns an inverse map from diffraction data to a complex real-space object, then trains it after passing the predicted object through known coherent diffractive imaging physics: overlap-based real-space consistency, the probe interaction, and far-field diffraction. The loss is applied in measurement space, so the network is optimized against the measured diffraction statistics rather than only against paired object targets.

This paper uses that physics-constrained formulation for CDI training. The measurement-consistency objective is not introduced here as a new CDI forward model. Instead, we train the same architecture families with and without the CDI forward model where both trainings were run. We then ask which architecture works best once the prediction is trained through the known diffraction physics.

3 Methods

3.1 Task-agnostic network formulation

Let x denote the task input and y the target field or reconstruction. We write the learned network as

$$\hat{y} = G_{\theta}(x). \quad (1)$$

For CDI, x is a stack of diffraction measurements and $\hat{y}$ is a complex object-patch representation, meaning local crops of the reconstructed complex object represented by real and imaginary channels. For BRDT, x is a Born-derived image-space initialization and $\hat{y}$ is the recovered scattering potential. For PDEBench CNS, x is a temporal history of physical fields and $\hat{y}$ is the predicted next state. The tasks use different input representations and objective functions, but share the same architectural principle: G_{θ} combines global spectral mixing with local convolutional feature recovery.

We decompose the network as

$$G_\theta = D_\theta \circ B_\theta \circ E_\theta, \quad (2)$$

where E_θ is the spectral-local encoder, B_θ is a ResNet bottleneck at reduced spatial resolution, and D_θ is a convolutional upsampling decoder.

We use subscripts to distinguish architectural positions rather than tensor dimensions. The index ℓ denotes the encoder block that maps z_ℓ to $z_{\ell+1}$; j indexes the six residual bottleneck blocks; s indexes resolution levels used for encoder-decoder skip connections; and k indexes Fourier modes. The retained Fourier-mode set is K , with $m = |K|$ in the architecture diagram. Superscripts such as $B_j^{\times 6}$ denote repeated blocks rather than exponentiation. Task subscripts such as CDI and CNS label task-specific outputs.

3.2 Spectral-local encoding

For a feature tensor z_ℓ , a spectral-local encoder block has the form

$$z_{\ell+1} = \sigma \left(L_\ell z_\ell + \mathcal{F}^{-1} [R_\ell(k) \mathcal{T}_K (\mathcal{F}(z_\ell)) (k)] \right), \quad (3)$$

where L_ℓ is a learned local convolutional map, $\mathcal{F}$ is the 2D Fourier transform, $\mathcal{T}_K$ retains a fixed set of Fourier modes, $R_\ell(k)$ is a learned complex spectral mixing tensor over retained modes, and σ is the GELU nonlinearity used in the implemented SRU-Net. In residual form, this can be written as

$$z_{\ell+1} = z_\ell + \sigma \left(L_\ell z_\ell + S_\ell z_\ell \right), \quad (4)$$

with S_ℓ denoting the truncated spectral operator. The local path preserves high-frequency and edge-like information that may be weakened by spectral truncation, while the spectral path provides global coupling.

3.3 Spectral-convolutional residual architecture

The architecture separates three roles. Spectral mixing in the encoder couples information across the full measurement field, where global aggregation is most useful for the inverse transform. The bottleneck then processes the resulting representation with local residual blocks at reduced spatial resolution; the architecture ablations in Section 5.3 test whether further spectral mixing in the bottleneck helps or hurts. Finally, the convolutional decoder synthesizes the real-space output, and the same ablations test whether it can be replaced with a lighter linear stage.

The implemented SRU-Net consists of an initial convolution, four spectral-local encoder blocks, two downsampling stages, a six-block residual bottleneck at $N/4$ spatial resolution, and a convolutional upsampling decoder. Figure 1 summarizes the data flow, and Table 1 lists the architectural choices that determine the shape of the model.

In the architecture diagram, ϕ_0 is the initial channel projection, E_0 through E_3 are encoder blocks, B_j denotes a residual bottleneck block, and U_1, U_2 are decoder upsampling stages. The spatial resolution is denoted by N , C_{in} is the task-specific input channel count, m is the number of retained Fourier modes, and e_s, d_s are encoder and decoder features at scale s . The skip projection P_s maps encoder features into the decoder channel space before additive fusion. Within encoder block E_ℓ , z_ℓ is the input feature tensor, $S_\ell = \mathcal{F}^{-1} R_\ell \mathcal{T}_K \mathcal{F}$ is the truncated spectral operator, L_ℓ is the local convolutional path, and σ is GELU. Here $\mathcal{F}$ is the Fourier transform, $\mathcal{T}_K$ keeps the retained modes, and R_ℓ mixes those modes. Within bottleneck block B_j , u_j is the bottleneck feature tensor, $R_j = \text{Conv}_{3 \times 3} \cdot \text{InstNorm} \cdot \text{GELU} \cdot \text{Conv}_{3 \times 3} \cdot \text{InstNorm}$ is applied before the residual addition, and α_j is its learned residual scale; the centered dot lists operations in left-to-right order.

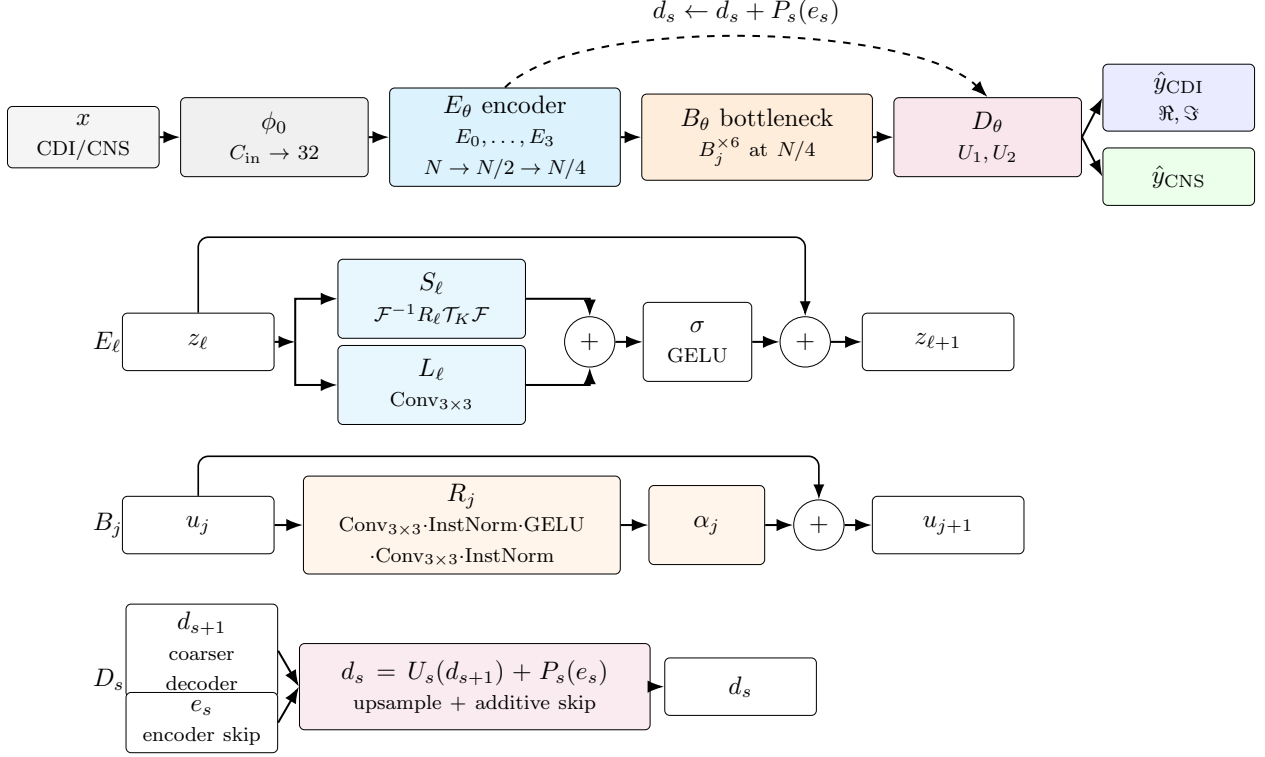


Figure 1: **SRU-Net combines spectral global mixing with local spatial processing.** Top: x is the task input, ϕ_0 projects C_{in} input channels to width 32, E_θ is the encoder over resolutions $N \rightarrow N/2 \rightarrow N/4$, B_θ is the six-block bottleneck, D_θ is the decoder, and $\hat{y}_{\text{CDI}}, \hat{y}_{\text{CNS}}$ are task-specific outputs. Middle: encoder block E_ℓ maps z_ℓ to $z_{\ell+1}$ using spectral path $S_\ell = \mathcal{F}^{-1} R_\ell \mathcal{T}_K \mathcal{F}$, local convolution path L_ℓ , GELU nonlinearity σ , and a residual bypass. The bottleneck detail shows that bottleneck block B_j maps u_j to u_{j+1} by applying R_j , then scaling it by α_j before adding it to the bypassed input. The R_j box uses centered dots to list left-to-right sequential application; $\text{Conv}_{3 \times 3}$ denotes a 3-by-3 convolution and InstNorm denotes instance normalization. The decoder detail writes the stage directly as $d_s = U_s(d_{s+1}) + P_s(e_s)$, where U_s upsamples the coarser decoder feature and P_s projects the encoder skip feature before additive fusion.

Here $\text{Conv}_{3 \times 3}$ denotes a 3-by-3 convolution. Within decoder stage D_s , U_s upsamples the coarser decoder state and P_s projects the matching encoder feature before fusion.

When skip connections are enabled, encoder features e_s are fused into decoder states d_s . For additive fusion,

$$d_s = U_s(d_{s+1}) + P_s(e_s), \quad (5)$$

where U_s upsamples the coarser decoder state and P_s projects the encoder feature into the decoder channel space. For concatenative fusion,

$$d_s = P_s([U_s(d_{s+1}), e_s]). \quad (6)$$

The CDI network outputs complex object patches in real/imaginary format. In BRDT, the same reconstruction architecture outputs a single real scattering potential. In the PDEBench experiments, the architecture is evaluated on real-valued physical field histories and next-state targets rather than diffraction measurements.

Table 1: **SRU-Net architecture summary.** The same architecture is used for CDI reconstruction and CNS field prediction, with task-specific input representations, output variables, and objective functions.

Component	Implementation details
Input representation	CDI uses grouped object-patch channels and returns complex real/imaginary patches. CNS uses real-valued field histories as input and predicts the next density, velocity, and pressure fields. In both settings, the first convolution maps the input to width 32.
Encoder	Four spectral-local blocks. Each block adds a learned Fourier-mode mixing branch and a reflected-padding 3×3 convolution branch inside an outer residual update; the branch sum is passed through GELU before the outer residual addition. The retained-mode count m is a configurable parameter; the default setting uses $m = 12$ unless otherwise stated. The encoder downsamples after the first two blocks, giving $N \rightarrow N/2 \rightarrow N/4$ and widths 32, 64, and 128.
Bottleneck	Six constant-resolution residual blocks at $N/4$. Each block uses a reflected-padding 3×3 convolution, instance normalization, GELU, a second reflected-padding 3×3 convolution, instance normalization, and a learned residual scale. The spectral-residual variant replaces this local bottleneck with residual blocks that add a shared factorized spectral branch.
Decoder and skips	The CDI path uses two transposed-convolution upsampling stages. The CNS model uses two pixel-shuffle upsampling stages and projected additive encoder-decoder skips from the $N/2$ and N encoder resolutions.
Output and loss	CDI predicts real/imaginary complex patches and trains through the coherent diffractive imaging consistency and diffraction operators. BRDT predicts a real scattering potential and trains with supervised object loss plus Born consistency. CNS predicts real field channels and trains against the next physical state.

3.4 CDI physics-consistency objective

For CDI, SRU-Net predicts local complex object patches,

$$\hat{o} = G_{\theta}(x). \quad (7)$$

The prediction is not trained only by comparing $\hat{o}$ to a paired object target. Instead, it is passed through a real-space consistency operator F_c , which stitches overlapping local patches into a common object canvas and re-extracts position-consistent patches. The consistent object patches are then passed through a known diffraction map F_d :

$$\hat{x} = F_d(F_c(\hat{o}; r); P), \quad (8)$$

where r denotes scan positions and P is the probe. The CDI objective is defined in measurement space:

$$\mathcal{L}_{\text{CDI}} = \mathcal{D}_{\text{amp}}(x, F_d(F_c(G_{\theta}(x); r); P)). \quad (9)$$

The reported CDI runs use detector-amplitude mean absolute error. The stored detector tensors are diffraction amplitudes, so after applying detector scaling the loss is

$$\mathcal{D}_{\text{amp}}(x, \hat{x}) = \frac{1}{\bar{x} + \epsilon} \frac{1}{|\Omega|} \sum_{i \in \Omega} |\hat{x}_i - x_i|, \quad (10)$$

where x_i and $\hat{x}_i$ are measured and predicted detector amplitudes, $\bar{x}$ is the mean measured detector amplitude in the batch, Ω indexes detector pixels and scan positions, and ϵ prevents numerical singularities. This is the physics-consistency objective used in the CDI architecture comparison and in the supervised-loss comparisons.

3.5 CDI supervised objective

To isolate the role of the training loss, we also train models with a supervised object-space loss when the same architecture was also trained with the CDI forward model. These runs use the same input data, architecture, optimizer, number of epochs, and evaluation metrics as their physics-consistency-trained counterparts, but the loss is applied directly to the predicted object representation rather than after the CDI forward model. The main architecture comparison uses the shared physics-consistency objective.

The supervised objective has the form

$$\mathcal{L}_{\text{sup}} = \mathcal{D}_{\text{obj}}(G_{\theta}(x), y), \quad (11)$$

where y is the object target and $\mathcal{D}_{\text{obj}}$ is the object-space reconstruction loss used by the corresponding supervised run. Metrics are computed after the same evaluation alignment and post-processing used for physics-consistency-trained models.

3.6 Born–Rytov diffraction tomography

We also evaluate SRU-Net on a Born–Rytov diffraction tomography (BRDT) inverse-scattering task. The target is the physical scattering potential

$$q(x, z) = k_m^2 \left[\left(\frac{n(x, z)}{n_m} \right)^2 - 1 \right], \quad (12)$$

where $n(x, z)$ is the refractive-index field, n_m is the background medium index, and k_m is the in-medium wavenumber. Under the first Born approximation, each incident plane wave produces a scattered field whose far-field samples are linear measurements of the Fourier transform of q on the corresponding Ewald arc. We write this discrete forward map as

$$s = \mathcal{B}(q), \quad (13)$$

where $s \in \mathbb{R}^{A \times D \times 2}$ is the complex sinogram over illumination angles and detector frequencies, represented by real and imaginary channels.

The BRDT network input is a Born-derived image-space initialization x_{BRDT} computed from the measured sinogram. The model predicts a single-channel scattering potential,

$$\hat{q}_{\theta} = G_{\theta}(x_{\text{BRDT}}). \quad (14)$$

All neural models use the same input representation and the same differentiable Born forward operator during training.

The BRDT objective combines supervised reconstruction with measurement-space Born consistency:

$$\begin{aligned} \mathcal{L}_{\text{BRDT}} = & \lambda_{\text{img}} \|\hat{q}_{\theta}^{\text{norm}} - q^{\text{norm}}\|_1 + \lambda_{\text{abs}} \|\mathcal{B}(\hat{q}_{\theta}) - s\|_1 \\ & + \lambda_{\text{rel}} \frac{\|\mathcal{B}(\hat{q}_{\theta}) - s\|_2}{\|s\|_2 + \epsilon} + \lambda_{\text{tv}} \text{TV}(\hat{q}_{\theta}) + \lambda_{\text{pos}} \|\min(\hat{q}_{\theta}, 0)\|_2^2. \end{aligned} \quad (15)$$

The image loss is computed in the normalized target space used by the model. Before applying the Born operator, predictions are converted back to physical q units using training-set normalization statistics. This prevents the measurement-consistency term from being evaluated on normalized, dimensionless outputs. We use $\lambda_{\text{img}} = 1$, $\lambda_{\text{abs}} = 0.1$, $\lambda_{\text{rel}} = 0.1$, $\lambda_{\text{tv}} = 10^{-5}$, and $\lambda_{\text{pos}} = 10^{-4}$.

3.7 PDEBench CNS supervised prediction

For the CNS benchmark, the same architecture is evaluated as a supervised forward predictor. Let

$$u_t = [\rho_t, v_{x,t}, v_{y,t}, p_t] \quad (16)$$

be the density, velocity, and pressure state. Given a history length h , the input is

$$x_t = [u_{t-h}, \dots, u_{t-1}], \quad (17)$$

and the model predicts

$$\hat{u}_t = G_\theta(x_t). \quad (18)$$

The training objective is mean-squared error on normalized physical fields:

$$\mathcal{L}_{\text{CNS}} = \|\hat{u}_t - u_t\|_2^2. \quad (19)$$

Evaluation is performed after denormalization and reports root mean squared error (RMSE), normalized RMSE (nRMSE), relative L_2 , and Fourier-band RMSE over low, mid, and high radial-frequency bands. The high-frequency metric is included because shocks and localized vortical features are precisely where global spectral models may lose detail.

4 Experiments

The main benchmark is synthetic CDI. We first compare supervised object-space training with physics-consistency training, then compare architectures under the same physics-consistency objective. Born–Rytov diffraction tomography is an inverse-wave transfer test. PDEBench CNS is a forward-prediction setting used to contextualize the architecture against operator-learning baselines. The paper’s main claim is therefore physics-consistent CDI reconstruction, not a PDEBench record.

4.1 Synthetic CDI benchmark

The CDI reconstruction task uses 128-by-128 synthetically generated diffraction data. Each sample is generated from sharp line-like objects, a known probe, and simulated far-field diffraction measurements. The network predicts complex object patches, the CDI consistency and diffraction operators map the prediction back to detector space, and reconstruction quality is measured against held-out object amplitude and phase.

The CDI experiments have two parts. First, we compare direct supervised object-space training with physics-consistency training where both trainings were run. This tests whether the forward-model objective is necessary under the synthetic CDI protocol. Second, we compare CNN, FNO, FFNO, U-NO, and SRU-Net under the shared CDI physics-consistency objective. This tests which network works best once the loss is fixed.

The synthetic CDI comparison uses $N = 128$ object grids, two training objects, two held-out test objects, 10^9 photons per exposure, 40 training epochs, learning rate 2×10^{-4} , ReduceLROnPlateau

scheduling, and complex-valued real/imaginary network output. Phase metrics are computed after the same global alignment procedure for all models. Appendix Table 9 lists the configuration of each model; parameter counts are unique trainable parameters in the prediction model. Appendix Table 8 summarizes benchmark scope, derived sample counts, training recipes, and evaluation alignment.

The CDI benchmark contains two independent training object instances and two independent held-out object instances, with 1458 held-out object-patch/scan samples used to reconstruct the held-out object images. Because the held-out samples share the same object instances, probe, scan protocol, and simulation settings, we frame this benchmark as a controlled comparison of losses and architectures within a fixed synthetic distribution; broader object-class generalization and real experimental robustness would require additional object families and real data, which are out of scope here.

4.2 BRDT inverse scattering

The BRDT experiment uses 128×128 scattering-potential fields and simulated complex sinograms from a differentiable Born forward model with 64 illumination angles and 128 detector samples. We use it to test whether the CDI reconstruction network also works on a related wave inverse problem. We train SRU-Net and FFNO for 40 epochs on the same 2048/256/256 train/validation/test split, using Adam with learning rate 2×10^{-4} , batch size 16, seed 42, and ReduceLROnPlateau scheduling. Both models use the same Born-derived image initialization as input and the same supervised-plus-Born-consistency objective defined above.

4.3 PDEBench CNS

The CNS benchmark is PDEBench 2D compressible Navier-Stokes at native 128×128 resolution. It stresses the same architectural tradeoff in a different physical setting: coherent global flow and wave structure must be predicted while preserving localized shocks, vortices, and high-gradient field features.

The CNS data are the official 128-by-128 periodic compressible-flow training trajectories at Mach number $M = 1.0$, shear viscosity $\eta = 0.01$, and bulk viscosity $\zeta = 0.01$. The spatial grid has $\Delta x = \Delta y = 1/128 = 0.0078125$, the temporal spacing is $\Delta t = 0.05$, and the spatial boundary condition is periodic, meaning that values wrap around at the domain edges as on a torus. Each trajectory contains the fields density, v_x , v_y , and pressure on a 128×128 grid. Splits are trajectory-level, normalization is computed from the train split only, and metrics are evaluated on denormalized predictions.

5 Results

The CDI results answer two questions: whether physics-consistency is needed in the loss, and which architecture performs best under that shared objective. BRDT provides inverse-scattering transfer evidence; CNS provides cross-domain context for forward prediction.

Table 2: **Synthetic CDI supervised and physics-consistency training.** Each block compares physics-consistency and supervised training for the same architecture when both trainings were run. This comparison tests the training loss, not the architecture alone. Lower MAE is better; higher SSIM is better.

Objective	Amp MAE	Phase MAE	Amp SSIM	Phase SSIM
<i>FFNO</i>				
PINN	0.0820	0.1380	0.8903	0.9596
Supervised	0.3515	0.0661	0.2650	0.9015

Table 3: **Synthetic CDI architecture comparison under physics-consistency training.** All models use the same $N = 128$ line-pattern protocol and CDI physics-consistency objective.

Model	Amp MAE ↓	Phase MAE ↓	Amp MSE ↓	Phase MSE ↓	Amp SSIM ↑	Phase SSIM ↑
CNN	0.1232	0.7863	0.0237	1.1843	0.7326	0.2638
FNO	0.1248	0.1435	0.0240	0.0323	0.7409	0.9335
FFNO	0.0820	0.1380	0.0106	0.0305	0.8903	0.9596
SRU-Net	0.0269	0.0721	0.0012	0.0078	0.9881	0.9947
U-NO	0.0932	0.0683	0.0132	0.0073	0.8280	0.9569

5.1 CDI supervised and physics-consistency training

Table 2 tests whether the physics-consistency objective is needed.¹ The objective is necessary: switching FFNO from supervised object-space training to physics-consistency training cuts amplitude MAE fourfold ($0.35 \rightarrow 0.08$), and similar gains hold for every architecture trained both ways. Supervised training can improve some phase-error metrics while degrading amplitude reconstruction or phase structure, whereas physics-consistency training gives stronger amplitude recovery and phase SSIM across the table.

5.2 CDI architecture comparison

After fixing the CDI physics-consistency objective for all models in the table, architecture still drives a multiplicative gap. Table 3 compares local, spectral, and spectral-local models under the same $N = 128$ line-pattern protocol. SRU-Net cuts amplitude MAE to 0.027, a $3.0\times$ improvement over FFNO (0.082), $4.6\times$ over the CNN baseline (0.123), and the lowest amplitude MAE, MSE, and SSIM in the table. U-NO gives the lowest phase MAE and MSE, while SRU-Net gives the strongest phase SSIM; FFNO improves over the CNN and FNO rows but does not close the amplitude gap to SRU-Net.

5.3 CDI architecture ablations

Table 4 isolates which architectural choices drive SRU-Net’s CDI behavior. The ablations show three things. First, the convolutional decoder is non-negotiable: replacing it with a linear decoder drives amplitude MAE from 0.027 to 0.135, a $5\times$ regression, and phase MAE from 0.072 to 0.117. Second, replacing the residual bottleneck with FFNO-style spectral blocks worsens both metrics, so additional spectral capacity placed inside the bottleneck is harmful, not helpful. Third, between

¹We write “PINN” and “+PINN” for training through the PychoPINN-style CDI measurement-space physics-consistency objective; the label does not indicate a new PDE-residual PINN formulation.

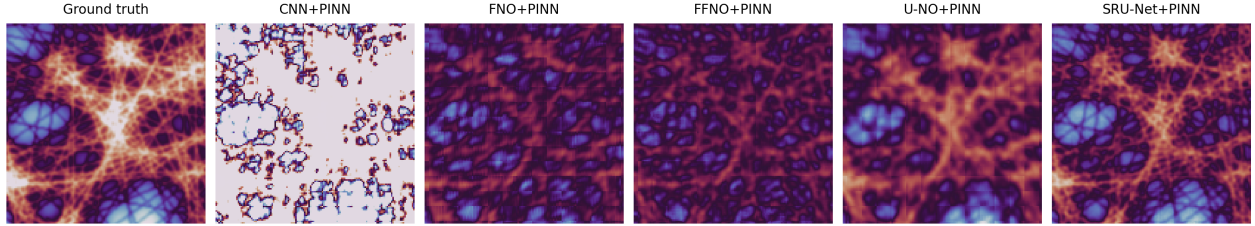


Figure 2: **Phase reconstruction comparison on synthetically generated CDI data.** The cropped comparison shows the target, CNN+PINN, FNO+PINN, FFNO+PINN, U-NO+PINN, and SRU-Net+PINN under the same $N = 128$ diffraction protocol. Each prediction is aligned to the target phase by a single global circular offset before display.

Table 4: **CDI spectral-component ablations.** Entries test whether SRU-Net’s performance is explained by spectral capacity alone or by the placement of spectral mixing inside an architecture with local convolutional paths, residual bottleneck processing, and convolutional upsampling. Lower MAE is better.

Model	Changed component	Amp MAE	Phase MAE
SRU-Net	none	0.0269	0.0721
SRU-Net with FFNO-style bottleneck	replace ResNet bottleneck	0.0312	0.0876
Spectral-bottleneck SRU-Net	spectral bottleneck	0.0249	0.0929
Spectral-bottleneck SRU-Net, one downsampling stage	shallower encoder	0.0355	0.0644
Spectral-bottleneck SRU-Net, linear decoder	lighter decoder	0.1347	0.1169

the reference SRU-Net and the spectral-bottleneck variants, amplitude and phase trade off along a Pareto front: the spectral bottleneck improves amplitude MAE (0.025 vs. 0.027) but worsens phase MAE (0.093 vs. 0.072), and shrinking to one downsampling stage flips this: phase improves (0.064) but amplitude regresses (0.036). The reference configuration is the only entry that is good on both metrics simultaneously.

5.4 BRDT inverse scattering

BRDT evaluates inverse scattering from simulated wave measurements. The target is the physical scattering potential q , and the input is a Born-derived image-space initialization computed from the measured complex sinogram. We train SRU-Net and FFNO on the same 2048/256/256 train/validation/test split for 40 epochs using the supervised object loss plus Born-consistency objective defined in Section 3.6. The model-based Born inverse is included as a non-learned reference.

Table 5 shows the BRDT comparison. SRU-Net reaches relative $L_2(q) = 0.288$ against 0.451 for FFNO and 0.380 for the model-based Born inverse. It also improves PSNR and SSIM relative to both references. Both learned models evaluate the 256-sample test split in under 0.7 seconds.

5.5 PDEBench CNS

Table 6 reports the CNS comparison. All models use five input states, 512/64/64 train/validation/test trajectories, 40 training epochs, MSE loss, and the same optimizer and normalization settings. FFNO gives the lowest aggregate and Fourier-band errors. SRU-Net is second by aggregate error. U-NO is nearly tied with FNO on aggregate error, but has lower mid- and high-frequency errors than FNO. U-Net is the weakest row in this comparison.

Table 5: **BRDT inverse-scattering comparison.** SRU-Net and FFNO are trained for 40 epochs with the same supervised object loss plus Born-consistency loss. The model-based Born inverse is included as a non-learned reference. Throughput is measured over the 256-sample test split on the same GPU. Lower error is better; higher PSNR, SSIM, and throughput are better.

Model	Procedure	Rel. $L_2(q)$	PSNR	SSIM	Params	Samples/s
Model-based Born inverse	iterative inverse	0.3796	28.23	0.9201	—	29.6
SRU-Net	supervised + Born	0.2876	30.64	0.9552	142k	375.9
FFNO	supervised + Born	0.4509	26.74	0.9120	27.4k	370.1

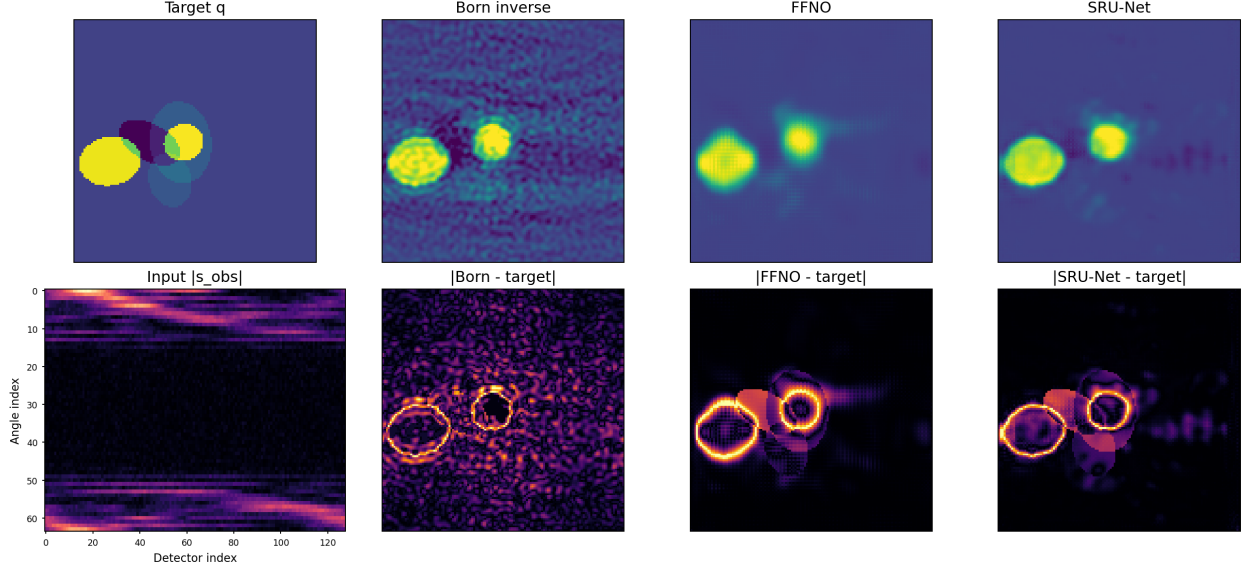


Figure 3: **BRDT representative example.** The figure separates the BRDT input $|s_{\text{obs}}(\theta, d)|$ from the target scattering potential q . The top row shows the target and three estimates $\hat{q}$; the bottom row shows the input sinogram magnitude followed by absolute target-domain errors under a shared error scale.

6 Discussion

Design rule. The CDI, BRDT, and CNS results, taken together, point to a simple rule for where spectral mixing belongs in a hybrid architecture. In wave inverse problems, the input lives in measurement space and the output lives in object space; the network’s job is to perform that domain change. Spectral coupling does its useful work *during* the inverse transform, after which the representation should be a local field that a convolutional decoder can synthesize. This predicts: spectral mixing in the encoder, local processing in the bottleneck, convolutional decoding. SRU-Net instantiates this and beats spectral-only and convolutional-only baselines on CDI by multiplicative margins. In forward PDE prediction, by contrast, input and output share a grid; there is no domain change to localize, and uniform spectral coupling at every layer is the right structural prior. FFNO instantiates this and remains the strongest model on CNS. The results therefore support an inverse-problem claim rather than a claim that SRU-Net should dominate forward prediction.

Generalization. BRDT fits the rule’s preconditions: Born-derived image-space input and real-space scattering-potential output. SRU-Net transfers cleanly. We would expect similar gains on

Table 6: **PDEBench CNS results.** All models use five input states, 512/64/64 train/validation/test split, and 40 training epochs. Low-, mid-, and high-band RMSE are Fourier-band errors. Lower values are better.

Model	nRMSE ↓	Low RMSE ↓	Mid RMSE ↓	High RMSE ↓
FFNO	0.0198	1.130	0.042	0.102
SRU-Net	0.0331	1.854	0.171	0.262
FNO	0.0384	2.134	0.122	0.433
U-NO	0.0385	2.190	0.079	0.261
U-Net	0.5386	30.988	0.645	1.743

other wave inverse problems with the same global-to-local domain structure, such as far-field phase retrieval without scanning, and treat this as a hypothesis the present results support rather than a theorem.

Ablations. Section 5.3 is a Pareto, not a sweep where the reference configuration dominates every cell. Replacing the convolutional decoder with a linear one is a hard failure ($5\times$ amplitude regression), so the decoder is a structural requirement. Between the local and spectral bottleneck, amplitude and phase trade off; the reference SRU-Net is the only configuration that is good on both.

Capacity. SRU-Net carries more parameters than FFNO on CDI (Appendix Table 9), and it is reasonable to ask whether the gap reduces to capacity. The CDI table itself argues against this: the CNN baseline at 4.66M parameters is the second-largest model in the comparison and the worst amplitude reconstructor (Amp MAE 0.123), and U-NO at intermediate capacity also does not approach SRU-Net. On BRDT, SRU-Net also has more parameters than FFNO, so that comparison is not parameter-matched. The spectral-component ablations vary capacity in both directions without reproducing the reference configuration’s joint amplitude–phase behavior. We therefore do not interpret the CDI gap as a parameter-count effect alone.

Scope. The CDI experiments use a controlled synthetic line-object distribution with two independent held-out object instances; derived object-patch/scan samples reconstruct those instances and are not independent object draws. Real experimental CDI data, broader object families, probe and scan perturbations, and lower-photon regimes are out of scope. The supervised-loss comparison only includes architectures trained both with and without the CDI forward model; supervised-loss claims are limited to those rows.

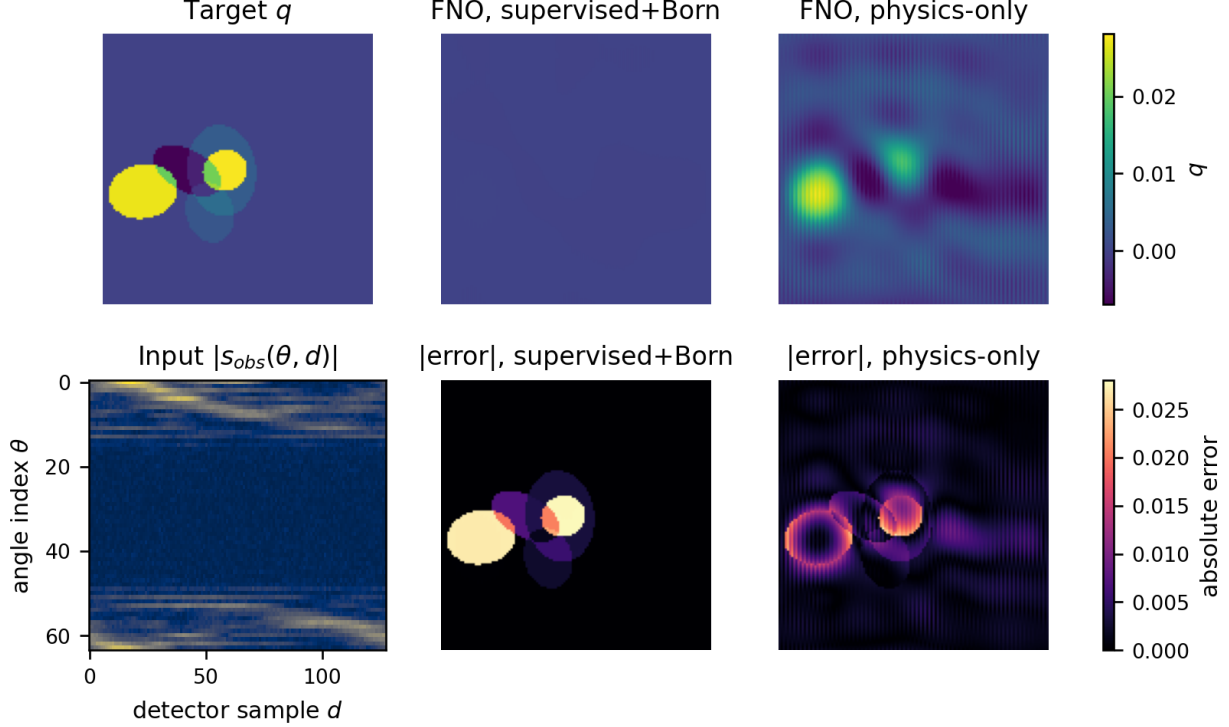


Figure 4: **BRDT FNO loss comparison.** Vanilla FNO did not converge under the supervised object loss plus Born-consistency objective used for the main BRDT comparison, producing a nearly constant reconstruction. Under a physics-only objective on the same BRDT split and representative example, the same architecture recovers visible object structure. We show this separately from Fig. 3, which compares the BRDT models used in Table 5.

A BRDT FNO Loss Comparison

B Additional CNS Visualization

C Additional CNS Comparisons

Table 7 reports additional SRU-Net comparisons on CNS. These results are not used as the main CNS comparison table. They show how input history, retained Fourier modes, and bottleneck depth affect aggregate and Fourier-band errors under fixed epoch counts. Spectral capacity alone does not determine performance; input history, local paths, decoder behavior, convergence, and parameter cost also matter.

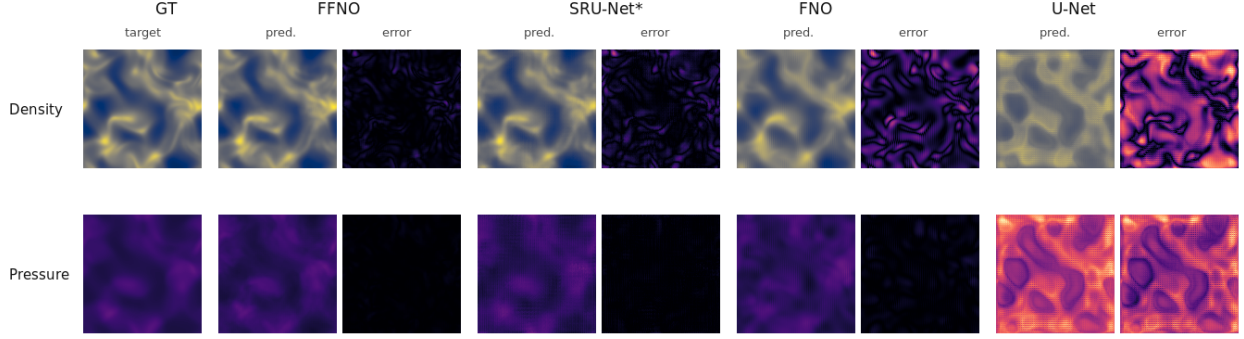


Figure 5: **PDEBench CNS fixed-sample predictions and errors.** The shared test sample comes from an earlier 2-input-time-step 512/64/64 CNS comparison and shows per-model error structure. Table 6 uses five input states, so this figure should not be read as the visual counterpart of the table.

Table 7: **Additional SRU-Net comparisons on PDEBench CNS.** The first column names the architecture or input-history change; the second column records the train/validation/test split and input history. All models use 40 training epochs. Low-, mid-, and high-band RMSE are Fourier-band errors. Lower values are better.

Tested change	Split and input	nRMSE ↓	Low RMSE ↓	Mid RMSE ↓	High RMSE ↓
SRU-Net, 2 input steps	512/64/64 split; 2 input steps	0.062	3.476	0.280	0.435
SRU-Net, 3 input steps	512/64/64 split; 3 input steps	0.046	2.560	0.216	0.347
SRU-Net, 32 Fourier modes	512/64/64 split; 2 input steps	0.065	3.676	0.242	0.347
Deeper shared-weight spectral bottleneck	1024/128/128 split; 2 input steps	0.045	2.483	0.217	0.294

D Benchmark Scope and Reproducibility

E Model Configurations

References

- [1] Oliver Hoidn, Aashwin Ananda Mishra, and Apurva Mehta. Physics constrained unsupervised deep learning for rapid, high resolution scanning coherent diffraction reconstruction. *Scientific Reports*, 13:22789, 2023. <https://www.nature.com/articles/s41598-023-48351-7>.
- [2] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. *International Conference on Learning Representations*, 2021. <https://arxiv.org/abs/2010.08895>.
- [3] Miguel Liu-Schiaffini, Julius Berner, Boris Boney, Thorsten Kurth, Kamyar Azizzadenesheli, and Anima Anandkumar. Neural operators with localized integral and differential kernels. *International Conference on Machine Learning*, 2024. <https://arxiv.org/abs/2402.16845>.

Table 8: **Benchmark scope and reproducibility details.** Independent instances refer to object instances for CDI, scattering-potential fields for BRDT, and trajectories for CNS. Patch and window counts are derived samples and should not be interpreted as independent object or trajectory draws.

Benchmark	Instances and derived samples	Training and objective	Evaluation alignment
Synthetic CDI	2 train objects and 2 held-out test objects; 8978 train and 1458 held-out object-patch/scan samples; metrics computed on held-out object reconstructions	single seed 3; 40 epochs; batch 16; Adam; learning rate 2×10^{-4} ; ReduceLROnPlateau; 10^9 photons; CDI measurement-space detector-amplitude MAE; supervised object-space controls	global phase alignment before phase metrics and display
BRDT	2048 train, 256 validation, and 256 test scattering-potential fields; field-level samples with no patch or window expansion	single seed 42; 40 epochs; batch 16; Adam; learning rate 2×10^{-4} ; ReduceLROnPlateau; supervised object loss plus Born-consistency terms	target-domain metrics in physical q units; representative figure uses a fixed held-out example
PDEBench CNS	512 train, 64 validation, and 64 test trajectories; 4096 train, 512 validation, and 512 test windows from the trajectory split	split seed 20260420; one run per row; 40 epochs; batch 4; Adam; learning rate 2×10^{-4} ; ReduceLROnPlateau; supervised one-step normalized-field MSE	denormalized metric evaluation; Fourier-band errors reported in Table 6

Table 9: **Model configurations by benchmark.** Parameter counts are unique trainable parameters in the prediction model. Duplicate wrapper names are counted once.

Benchmark	Model	Training	Width	Modes	Enc.	Bott.	Down	Unique params
Synthetic CDI	CNN	physics-consistency	—	—	—	—	—	4,661,212
Synthetic CDI	FNO	physics-consistency	32	12	4	—	—	636,450
Synthetic CDI	FFNO (pinn_ffno)	physics-consistency	32	12	4	—	—	124,966
Synthetic CDI	SRU-Net	physics-consistency	32	12	4	6	2	9,003,299
Synthetic CDI	U-NO	physics-consistency	32	12	4	—	—	1,257,858
Synthetic CDI	CNN	supervised	—	—	—	—	—	4,612,418
Synthetic CDI	FFNO (supervised_ffno)	supervised	32	12	4	—	—	124,966
Synthetic CDI	U-NO	supervised	32	12	4	—	—	1,257,858
PDEBench CNS	FFNO	supervised	64	16	24	—	—	1,074,440
PDEBench CNS	SRU-Net	supervised	32	12	4	6	2	8,395,270
PDEBench CNS	FNO	supervised	32	12	4	—	—	358,628
PDEBench CNS	U-NO	supervised	32	12	4	—	—	1,260,548
PDEBench CNS	U-Net	supervised	32	—	—	—	—	7,768,036
BRDT	SRU-Net	supervised + Born	16	8	2	2	1	142,018
BRDT	FFNO	supervised + Born	16	8	4	—	—	27,394

- [4] Gege Wen, Zongyi Li, Kamyar Azizzadenesheli, Anima Anandkumar, and Sally M. Benson. U-FNO: an enhanced Fourier neural operator-based deep-learning model for multiphase flow. *Advances in Water Resources*, 163:104180, 2022. <https://doi.org/10.1016/j.advwatres.2022.104180>.
- [5] Alasdair Tran, Alexander Mathews, Lexing Xie, and Cheng Soon Ong. Factorized Fourier neural operators. *International Conference on Learning Representations*, 2023. <https://arxiv.org/abs/2111.13802>.
- [6] Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflueger, and Mathias Niepert. PDEBench: An extensive benchmark for scientific machine learning. *Advances in Neural Information Processing Systems*, 2022. <https://arxiv.org/abs/2210.07182>.